

A Simple Perceptual Reference Method for Reducing False-Positive UAP Observations v1.09: The 'STAR Method' (Stable Thumb Alignment Reference) of Reducing False-Positive Observations

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Abstract

Reports of Unidentified Aerial Phenomena (UAP) frequently contain descriptions of erratic or “darting” motion that can often be traced to two distinct classes of false-positive mechanisms. The first arises from human perceptual processes, including the autokinetic effect [Sherif, 1935], microsaccades [Martinez-Conde et al., 2004], and fixational instability, all of which can cause stationary point-light sources to appear to move unpredictably in the night sky.

The second arises from smartphone imaging systems, where hand tremor, electronic image stabilization, rolling-shutter distortion, and multi-frame HDR processing can introduce artificial motion or duplicate light artifacts in recorded video.

This paper introduces the *STAR Method* (Stable Thumb Alignment Reference), a simple perceptual stabilization technique designed to help observers distinguish internal visual motion from genuine external object motion before attempting to record an event. The method uses the observer’s extended thumb as a nearby alignment reference against distant point-light sources, allowing rapid comparison between perceived motion and stable visual structure.

A two-stage false-positive model is proposed: Stage 1 involves perceptual distortions occurring prior to recording, while Stage 2 involves distortions introduced by the camera pipeline. By providing a stable visual reference, the *STAR Method* reduces the likelihood that observers will misinterpret perceptual illusions as anomalous motion, thereby preventing many misleading smartphone videos from being created.

The method is intentionally low-technology, requiring no specialized equipment or instrumentation, and is designed to be easily taught to the general public, civilian

observers, pilots, skywatchers, and UAP research organizations. It is further proposed that UAP-related reporting websites and observational organizations may benefit from incorporating the STAR Method and its accompanying visual reference graphic into public educational material intended to improve the reliability and evidentiary value of submitted observations.

The acronym “STAR” (Stable Thumb Alignment Reference) was intentionally selected to be memorable, intuitive, and uniquely searchable within internet search engines, allowing the method to be more easily referenced, located, and disseminated by both researchers and lay observers.

Version 1.07 includes an associated .ZIP archive containing images in Arabic, Chinese, Croatian, Czech, Danish (Denmark), English, Finland, French, German, Greek, Hebrew, Hindi, Hungarian, Italian, Japanese, Korean, Netherlands-Dutch, Norwegian, Polish, Portuguese, Romanian, Russian, Serbian, Spanish, Swedish and Turkish.

1 Introduction

Human visual perception is not a passive recording system, but an active interpretive process subject to a variety of physiological and environmental distortions. Under low-reference viewing conditions—particularly when observing isolated point-light sources in the night sky—stationary objects can appear to drift, dart, pulse, or accelerate despite possessing little or no actual motion. Such effects are well documented within perceptual psychology, aviation medicine, and visual neuroscience [Sherif, 1935, Clarke, 1960, Martinez-Conde et al., 2004, Engbert and Kliegl, 2003].

Within the context of Unidentified Aerial Phenomena (UAP) reporting, these perceptual effects become especially significant because observers frequently attempt to record unusual observations using handheld smartphone cameras. This transition from direct observation to digital recording introduces a second class of false-positive mechanisms involving camera shake, electronic stabilization artifacts, rolling shutter distortion, frame interpolation, and multi-frame HDR processing [Hamelin and Others, 2010, Liu and Others, 2009, Hedborg and Others, 2011, Hasinoff and Others, 2016]. As a result, perceptual illusions that originate within the observer can become amplified or transformed into apparently anomalous motion within recorded video.

After illustrating the relationship between microsaccades and the autokinetic Effect in the eye, this paper introduces the *STAR Method* (Stable Thumb Alignment Reference), a simple observational technique intended to help individuals distinguish internally generated apparent motion from genuine external object motion prior to recording an event. By using the observer’s thumb as a stable nearby alignment reference, the method provides a rapid perceptual comparison that may reduce the likelihood of misinterpreting autokinetic or microsaccadic effects as extraordinary aerial behavior.

Engbert and Mergenthaler note that “microsaccades are triggered dynamically” in response to low retinal image slip [Engbert and Mergenthaler, 2006, p. 7192], suggesting that perceptual stabilization mechanisms may contribute to apparent motion phenomena during fixation of distant point-light sources in the night sky.

The STAR Method is **intentionally low-technology** and **requires no specialized equipment**, making it suitable for broad public use, educational outreach, and possible

adoption by UAP reporting organizations seeking to improve the evidentiary quality of submitted observations.

This paper proposes a simple perceptual technique— *The 'STAR Method' (Stable Thumb Alignment Reference)*—to help observers determine whether perceived motion is internal or external before attempting to record the phenomenon. This method is intended to reduce the number of misleading smartphone videos submitted to UAP organizations.

2 Background

2.1 Perceptual Sources of False Motion

Human visual perception is susceptible to several well-documented illusions when observing isolated point sources in darkness. The autokinetic effect causes stationary lights to appear to move unpredictably [Sherif, 1935].

Recent research into fixational eye movements further emphasizes that “our eyes must produce miniature (‘fixational’) eye movements to counteract perceptual fading” [Engbert and Mergenthaler, 2006, p. 7192], supporting the possibility that internally generated stabilization processes may contribute to apparent motion perception under low-reference night-sky conditions.

Engbert and Kliegl further observed that “shifts of visual attention can produce a bias in this inherent randomness” [Engbert and Kliegl, 2003, p. 1043], suggesting that attentional processes themselves may influence the apparent motion characteristics perceived during prolonged fixation of distant point-light sources.

3 Related Work

False-positive observations of point light sources in low-light environments have been extensively studied across several scientific domains, including perceptual psychology, aviation safety, astronomy, and computational imaging. The present work builds upon this interdisciplinary foundation by integrating findings from human visual perception with known limitations of smartphone imaging systems.

Fixational eye movements, including drift, tremor, and microsaccades, represent normal neurological stabilization processes that occur continuously during attempted steady fixation [Leigh and Zee, 1999].

3.1 Perceptual Illusions and Apparent Motion

The autokinetic effect, first systematically described by Sherif [1935], demonstrates that a stationary point of light viewed against a featureless background can appear to move unpredictably. Subsequent research in aviation psychology confirmed the operational significance of this illusion, particularly for pilots conducting night operations [Clarke, 1960, Federal Aviation Administration, 2016]. These findings establish that perceived motion in the night sky can arise from internal visual processes rather than external object dynamics.

Microsaccades and fixational eye movements provide a physiological basis for such illusions. [Martinez-Conde et al. \[2004\]](#) and [Engbert and Kliegl \[2003\]](#) show that involuntary microsaccades introduce apparent motion during attempted fixation, especially when observing isolated stimuli. These effects are amplified in low-contrast environments, making them directly relevant to UAP observations. The broader neurophysiological context is well documented in [Leigh and Zee \[2015\]](#).

3.2 Relationship Between Microsaccades and the Autokinetic Effect

Microsaccades are involuntary, extremely small eye movements that occur approximately 1 - 2 times per second and are typically only a few arc-minutes in amplitude. These movements are continuously present in normal human vision and serve an important physiological role in preventing perceptual fading caused by retinal adaptation. If retinal images were held perfectly stationary, visual perception would gradually diminish within seconds. Microsaccades continuously refresh the retinal image and help maintain visual perception.

The autokinetic effect, also known as autokinesis, is a visual illusion in which a stationary point of light viewed in darkness or against a featureless background appears to move. Under such conditions, observers may perceive the light as drifting, darting, accelerating, or changing direction despite the absence of actual object motion. Reports of this phenomenon have historically been common in aviation and nocturnal observation environments.

The association between microsaccades and the autokinetic effect can be summarized as follows:

1. An observer fixates on a single star, planet, aircraft light, or distant ground light under dark viewing conditions.
2. The absence of surrounding visual reference points reduces the availability of a stable spatial frame of reference.
3. Microsaccades produce subtle shifts of the light image across the retina. Under ordinary viewing conditions, these shifts are normally not consciously perceived.
4. In low-reference visual environments, the visual system may attribute these retinal image displacements to external object motion rather than to involuntary eye movement.
5. As a result, a stationary light source may appear to “drift,” “zig-zag,” “dart,” or maneuver unpredictably despite remaining physically stationary.

3.3 Astronomical Misidentification and Atmospheric Effects

Astronomical literature has long recognized that stars and other point sources can appear to “dance” or “jump” due to atmospheric scintillation. [Dravins and Others \[1997\]](#) provide a detailed characterization of intensity fluctuations caused by turbulence, which can mimic erratic motion. These effects contribute to misidentification of celestial objects as anomalous aerial phenomena.

3.4 Camera-Induced Motion Artifacts

A second body of related work concerns the behavior of digital imaging systems under low-light and high-zoom conditions. Hand tremor has been shown to produce amplified apparent motion in video recordings, particularly when imaging distant point sources [Hamelin and Others, 2010]. Modern smartphones attempt to compensate for such motion using electronic image stabilization (EIS), but stabilization algorithms can introduce additional artifacts, including warping, frame-to-frame jitter, and false lateral motion [Liu and Others, 2009].

Neither consumer smartphone camera hardware nor associated image-processing software can match the sophistication, sensor stability, optical precision, or computational correction systems typically present in professional imaging equipment. As a result, amateur recordings frequently contain visual artifacts that may be misinterpreted as unusual object behavior.

Under low-light conditions in particular, smartphone cameras may produce motion smearing, rolling-shutter distortions, compression artifacts, exposure instability, frame blending, or “ghost” trailing effects.

These artifacts can cause an object to appear semi-transparent, elongated, amorphous, or unstable in shape, creating the false impression that the recorded object is partially formed, fluctuating, or exhibiting anomalous motion despite the effect originating entirely from the imaging system itself.

Rolling shutter distortion, inherent to CMOS sensors, further contributes to false motion by reading out the image line-by-line rather than simultaneously. This can cause skew, wobble, and apparent acceleration of bright point sources [Hedborg and Others, 2011]. Multi-frame HDR and burst-processing pipelines, now standard in mobile devices, may also produce ghosting or duplicate images when bright lights move relative to the sensor during frame fusion [Debevec and Others, 2008, Hasinoff and Others, 2016].

Because smartphone cameras possess a significant potential to introduce visual distortions and imaging artifacts, it is essential that observers first determine whether the perceived motion or anomalous appearance originates from genuine external behavior or from limitations inherent to the imaging system itself.

3.5 Human Factors in Anomalous Observation Reporting

Studies in aviation safety, perceptual psychology, and anomalous event reporting have consistently emphasized the importance of structured observational methodology and reliable visual reference systems. Haines [2001] note that perceptual limitations, environmental conditions, and imaging artifacts may contribute significantly to the misinterpretation of aerial phenomena, and recommend improved observer awareness, training, and methodological rigor during the observation process.

The *STAR Method* (Stable Thumb Alignment Reference), presented at the conclusion of this paper, is conceptually consistent with these recommendations. By providing the observer with a simple, continuously available nearby alignment reference, the method introduces an immediate perceptual stability control intended to help distinguish internally generated apparent motion from genuine external object movement prior to recording or reporting an event.

3.6 Historical Pilot Misinterpretations Associated with Microsaccades and the Autokinetic Effect

A significant portion of the historical literature concerning autokinesis and microsaccadic visual effects has focused on aviation environments and pilot observations in such books as Beaty's *The Naked Pilot* [Beaty, 1995] and Hawkins' *Human Factors in Flight* [Hawkins, 1993]. This emphasis has occasionally created the misleading impression that such perceptual errors are unique to pilots or to aircraft cockpits.

In reality, the underlying physiological mechanisms are fundamental properties of normal human vision and are therefore not exclusive to aviation personnel. Microsaccades, retinal image instability, and the resulting autokinetic illusion arise from normal visual processing under low-reference viewing conditions. Any observer viewing isolated points of light in darkness or against a featureless background may experience similar perceptual distortions.

The aviation environment merely provided an early and highly documented context in which these visual phenomena became operationally significant due to the frequency of nocturnal observations and the safety-critical nature of flight operations.

Consequently, the perceptual mechanisms associated with autokinesis should not be regarded as pilot-specific visual abnormalities, but rather as general characteristics of human visual perception that can affect any observer under appropriate environmental conditions.

3.7 The Importance of Obtaining a Strong, Stable Reference Point Within the Field of View

One of the most important factors in accurately evaluating apparent aerial motion is the presence of a stable visual reference point within the observer's field of view. Without nearby environmental references, the human visual system may have difficulty determining whether perceived motion originates from the observed object itself, from involuntary eye movement, or from movement of the observer.

Dark skies, isolated light sources, atmospheric conditions, and featureless backgrounds substantially reduce the availability of spatial reference information. Under such conditions, normal visual processes such as microsaccades, fixation instability, and perceptual drift may contribute to the false appearance of object motion. The observer may therefore perceive stationary or slowly moving objects as rapidly maneuvering, accelerating, changing direction, or hovering unpredictably.

The inclusion of a stable foreground or background reference point allows the observer to compare the apparent position of the object against a fixed environmental structure. Trees, buildings, rooftops, power poles, mountain ridges, or intentionally introduced nearby alignment references can significantly improve perceptual stability and reduce the likelihood of autokinetic misinterpretation.

For this reason, obtaining and maintaining a strong visual reference point should be considered an essential component of reliable nighttime observation and recording methodology, particularly when attempting to distinguish genuine external motion from internally generated perceptual effects or camera-induced artifacts.

4 The '*STAR Method*' (*Stable Thumb Alignment Reference*)

For rapid field use, the *STAR Method* can be reduced to a short sequence of simple steps. These steps are intended for observers who need an immediate way to determine whether perceived motion is likely due to internal visual processes such as the autokinetic effect or microsaccades.

The FAA notes that autokinesis occurs because “motion perception is always relative to some fixed reference point,” and warns that isolated lights against dark backgrounds may appear to move despite remaining stationary. These effects are sufficiently well established to represent recognized aviation safety hazards during nighttime flight operations. The STAR Method specifically attempts to introduce an immediate nearby alignment reference in order to reduce ambiguity arising from reference-free visual conditions [Hartman \[2025\]](#).

1. Extend your arm fully and raise your thumb into your line of sight near the light, positioning the light just above your thumb (orange star in Fig. 1).
2. Brace your arm with your other hand, forming a stable triangular structure.
3. While looking at the light, briefly shift your gaze so that both the light and your thumb remain visible.
4. Compare the apparent motion of the object relative to your thumb.
5. Observe whether the light appears to move independently of the thumb.
6. If the apparent motion changes, diminishes, or follows your eye movement, classify the effect as likely perceptual in origin.
7. If the motion remains stable and independent of eye movement or thumb reference, consider documenting the event for further observation.



Figure 1: The 'STAR Method' used as a stable perceptual alignment reference against distant point-light sources.

This version provides a fast, low-effort check that can be performed before attempting to record the phenomenon, reducing the likelihood of false-positive smartphone videos.

4.1 Summary

Together, these bodies of work demonstrate that both perceptual and technological mechanisms can generate false motion in UAP observations. The present contribution synthesizes these findings into a practical, field-deployable method designed to reduce the incidence of such false positives.

5 Conclusion

False-positive UAP observations arise from a combination of perceptual and technological factors. Perceptual illusions such as the autokinetic effect, microsaccades, and fixational instability can cause stationary point-light sources to appear to move erratically, while smartphone imaging systems introduce additional distortions through hand tremor, electronic stabilization, rolling-shutter readout, and multi-frame processing. Together, these mechanisms create a two-stage false-positive cascade in which an observer first misinterprets a visual illusion as anomalous motion and then unintentionally amplifies that illusion by attempting to record it.

The '*STAR Method*' provides a simple, low-effort intervention at the earliest point in this cascade. By giving observers a stable visual reference, the method helps them determine whether perceived motion is internal or external before they reach for a smartphone. If widely adopted by amateur observers, this technique would substantially reduce the number of misleading videos submitted to UAP organizations. Fewer false-positive recordings would improve the overall signal-to-noise ratio of public reports, allowing investigators to focus their efforts on cases with genuine evidentiary value.

Because the method requires no equipment, training, or specialized knowledge, it is well suited for dissemination through citizen-science networks, public outreach materials, and UAP reporting platforms. Incorporating the '*STAR Method*' into standard witness instructions would represent a practical, scientifically grounded step toward improving the reliability and interpretability of amateur UAP observations.

6 A Unifying Perspective

Although the '*STAR Method*' is a practical observational tool rather than a theoretical framework, it nonetheless unifies two domains that are often treated separately in UAP research. On one side are the internal mechanisms of human visual perception, including autokinesis, microsaccades, and fixational instability. On the other side are the external mechanisms of digital imaging systems, such as electronic stabilization, rolling-shutter readout, and multi-frame processing.

These two domains—perception and instrumentation—are typically analyzed independently, yet both contribute to the same class of false-positive UAP observations. The '*STAR Method*' brings these elements together by providing a single, simple intervention that addresses the earliest point of failure common to both: the observer's initial interpretation of motion. In this sense, the method serves as a small but meaningful unifying bridge between human factors research and the technical behavior of modern imaging devices.

It is the view of the author that improving an observer's ability to recognize and eliminate perceptual misinterpretations prior to recording may substantially reduce the overall number of false-positive UAP reports and associated observational errors.

By framing the problem in this integrated way, the method aligns with broader efforts to create coherent, cross-disciplinary approaches within UAP research. It demonstrates that improving observational reliability requires understanding how perceptual and technological processes interact, rather than treating them as isolated contributors.

7 Author’s Note on Scope

Although this work lies outside the author’s usual domain of theoretical research, it is presented here because of its practical value to observers and investigators. The ‘*STAR Method*’ is not part of the author’s broader UNIFICATION frameworks, nor does it attempt to advance any theoretical claims. Instead, it offers a simple, empirically grounded guideline intended to reduce preventable false-positive UAP observations.

The motivation for including this method is straightforward: many misleading reports and smartphone videos arise from well-understood perceptual and imaging artifacts. A small, accessible intervention that helps observers distinguish internal visual effects from genuine external motion has the potential to improve the overall quality of public UAP data. For this reason, the author considers the method worth documenting and sharing, despite its methodological rather than theoretical nature.

More broadly, widespread use of this method among amateur observers would have a cumulative effect on the quality of UAP reporting. Most UAP data originates not from specialized instruments but from members of the public who encounter unexpected lights in the sky. A simple perceptual check that can be performed in seconds has the potential to reduce thousands of false-positive reports over time. By encouraging observers to verify perceived motion before recording, the method helps ensure that only genuinely anomalous events progress to the documentation stage. This shift would not only reduce analytical burden on research organizations but also increase the proportion of cases with meaningful evidentiary value. In this way, a small individual action scales into a significant community-level improvement in observational reliability.

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